Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies:**
  - Data Collection with API and Web Scraping
  - Exploratory Data Analysis with Data Visualization and SQL
  - Interactive Map with Folium
  - Dashboards with Plotly Dash
  - Predictive Analysis
- **Summary of all results:**
  - Exploratory Data Analysis results
  - Interactive maps and dashboards
  - Predictive results

Introduction

Background:

- The objective of this project is to predict if the Falcon 9 first attempt will be landed successfully. As per SpaceX website Falcon 9 rocket launch cost $62 million while other providers cost more than $165 million each. By determining if the stage will land, we can determine the cost of a launch. This information could provide insight to other competing companies that provides the same services.

Issues and Questions:

- What are features of a successful or failed landing?
- What are the effects of each relationship of the rocket variable on the success or failure landing?
- What are the variable that will allow spaceX to achieve best landing success rate?

Section 1

Methodology



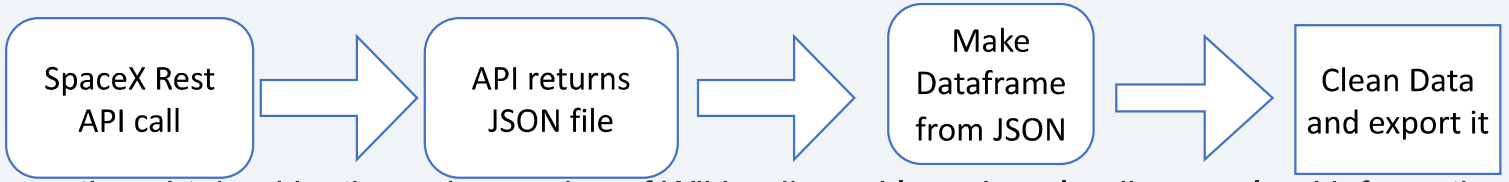
Methodology

Executive Summary

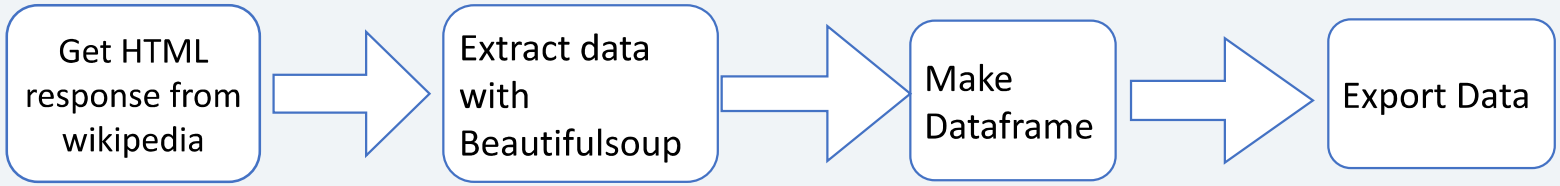
- Data collection methodology:
  - SpaceX Api: https://api.spacexdata.com/v4/rockets/
  - Web Scraping data from wiki: "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"
- Perform data wrangling:
  - Collected data was enriched by creating a landing outcome label based on outcome data after summarizing and analyzing features.
- Perform exploratory data analysis (EDA) using visualization and SQL:
  - Visualize variable relationships and look at the data in aggregate.
- Perform interactive visual analytics using Folium and Plotly Dash:
  - Mark all launch sites on a map
  - Mark successful and failed launches
  - Calculate distance to proximate locations
  - Provide for interactive exploration of the data
- Perform predictive analysis using classification models:
  - Data that was collected this step were normalized, divided in training and test data sets and evaluated by four different classification models, being the accuracy of each model evaluated using different combination of parameters.

Data Collection

- **Data sets collected from SpaceX API and web scraping Wikipedia**
  - The information obtained by the API and rocket, launches, payload information.
    - The Space X API URL is api.spacexdata.com/v4/



  - The information obtained by the web scraping of Wikipedia and launches, landing, payload information
    - URL: https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922



Data Collection - SpaceX API

1. Getting Response from API

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url)
```

2. Convert Response to JSON file

```
response = requests.get(static_json_url).json()

data = pd.json_normalize(response)
```

3. Transform data

```
getBoosterVersion(data)
getLaunchSite(data)
getPayloadData(data)
getCoreData(data)
```

4. Create dictionary with data

```
[34]:
launch_dict = {'FlightNumber': list(data['flight_number']),
'Date': list(data['date']),
'BoosterVersion':BoosterVersion,
'PayloadMass':PayloadMass,
'Orbit':Orbit,
'LaunchSite':LaunchSite,
'Outcome':Outcome,
'Flights':Flights,
'GridFins':GridFins,
'Reused':Reused,
'Legs':Legs,
'LandingPad':LandingPad,
'Block':Block,
'ReusedCount':ReusedCount,
'Serial':Serial,
'Longitude': Longitude,
'Latitude': Latitude}
```

5. Create Dataframe

```
data=pd.DataFrame.from_dict(laucnh_dicy)
```

6.Filter dataframe

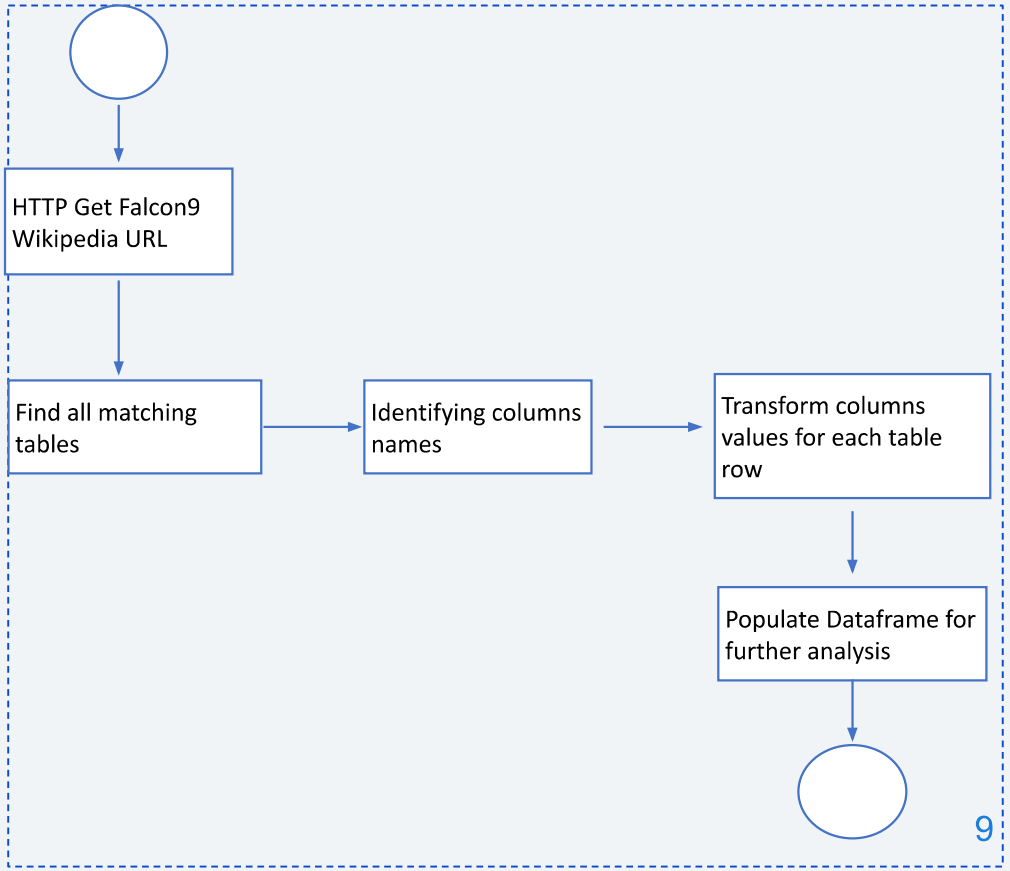
```
data_falcon9 = data[dara['RoosterVersion']!='Falcon1']
```

7. Export to file

```
data_falcon9.to_csv('dataset_part_1.csv', index=False)
```

Data Collection – Scraping

- Web scraping workflow:
  - Send HTTP Get to Falcon 9 Launch page at Wikipedia, to retrieve the HTML source
  - Create a BeautifulSoup object to extract the tables containing launch data
  - Populate a Pandas DataFrame using the extracted columns
- Github URL



Data Wrangling

- **Goals:**
  - o Use Exploratory Data Analysis (EDA) to find patterns in the data.
  - o Determine the labels for training supervised models.
- **Steps:**
  - o Find missing values as a percentage of each attribute
  - o Identify column types, numerical or categorical
  - o Show launches per Launch Site
  - o Show distribution of Orbit type in the data set
  - o Explore outcomes, and group them by binary outcome (success or failure)
  - o Store outcome as "Class" label, to be used as target value in training
- Github URL

EDA with Data Visualization

- EDA includes visually exploring and summarizing the main attribute of a Data Set. The objective is to comprehend the data distribution, identifying the patterns and find relationships between variables. The following are the visualized relationships to gain insight into the importance of each variable:
  - o Flight Number and Outcome
  - o Flight Number and Launch Site
  - o Payload and Launch Site
  - o Orbit and Outcome
  - o Flight Number and Orbit
  - o Payload and Orbit
  - o Yearly success rate
- Expand categorical variables into "dummy" columns
- Convert numerical columns into float64

EDA with SQL

- **SQL queries executed in order to gather and comprehend data from the dataset:**
  - Displays the names if the unique launch sites of the mission.
  - Displays 5 records where launch site names begin with string 'CCA'.
  - Display the total payload mass carried by boosters launched by 'NASA (CRS)'.
  - Display the average payload mass carried by the v1.1 Falcon 9 booster.
  - List the date of the first successful ground landing outcome.
  - List the booster versions with successful outcomes landing on the drone ship with payloads between 4000kg and 6000kg.
  - List the number of successful and failed mission outcomes.
  - List all of the booster versions that carried the max payload mass.
  - List the month name, outcome, booster version, and launch site for missions with failure outcomes landing on a drone ship in 2015.
  - Show the distribution of outcomes between June 4th, 2010 and March 20th, 2017.
- Github URL

Build an Interactive Map with Folium

- To find geographical patterns in the data the following items were marked on a map of launch sites:
  - All Launch Sites
  - Successful and Failed Launches
  - Distances between a launch site and proximate landmark
- Github URL

Build a Dashboard with Plotly Dash

- To enable interactive exploration of the data, a Plotly Dash dashboard was developed to include:
  - o A dropdown selector to choose a Launch Site, affecting:
    - ▪ A pie chart
      - • All sites selected: shows the breakdown of successful outcomes across all sites
      - • A launch site selected: shows the breakdown of successful vs failed launches for the given site
    - ▪ A scatter plot
      - • All sites selected: shows outcome by payload mass and booster version for all sites
      - • A launch site selected: shows outcome by payload mass and booster version for the given site
      - o A Payload Mass range selector that filters data points on the scatter plot

- Plotly Dash App (GitHub URL)

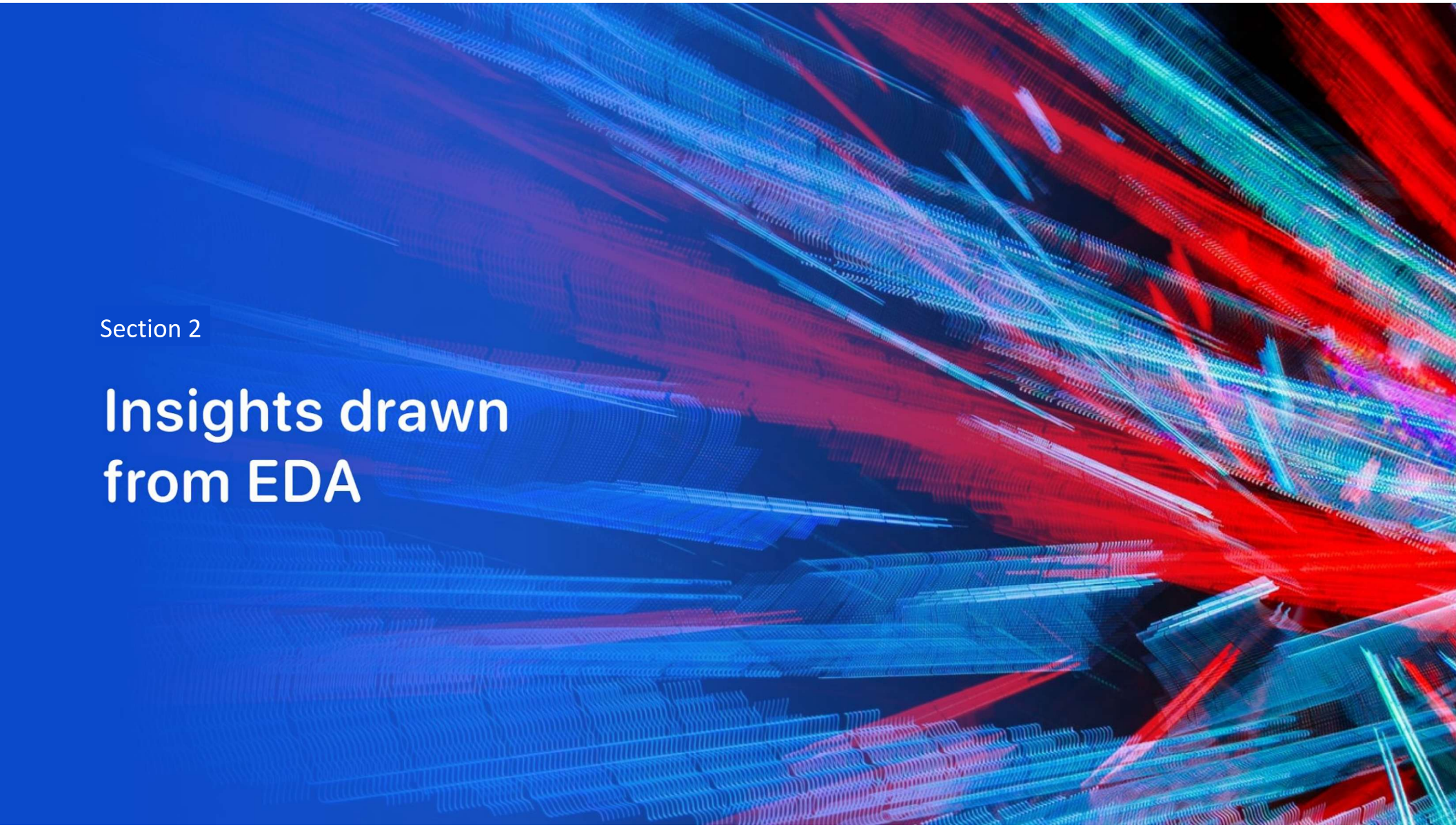
Predictive Analysis (Classification)

- Data preparation
  - Load dataset
  - Normalize data
  - Split data into training and tests sets
- Model preparation
  - Selecting different machine learning algorithms
  - Set parameters for each algorithm to GridSearchCV
  - Tune different parameters using GridSearchCV,
- Model evaluation
  - Use the best hyperparameters for each type of mode and compute accuracy for each model with test dataset
  - Plot Confusion Matrix
- Model Comparison
  - Compare model based on their accuracy and select the model according to accuracy result.

- Notebook(Github URL)

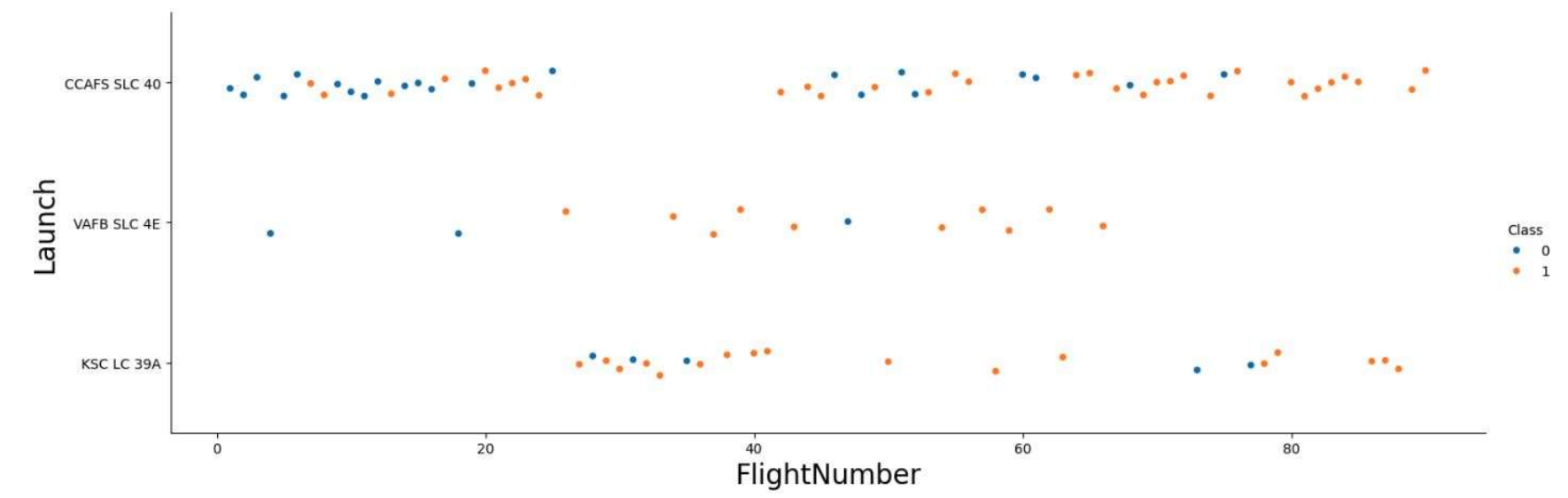
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results


Section 2
Insights drawn
from EDA

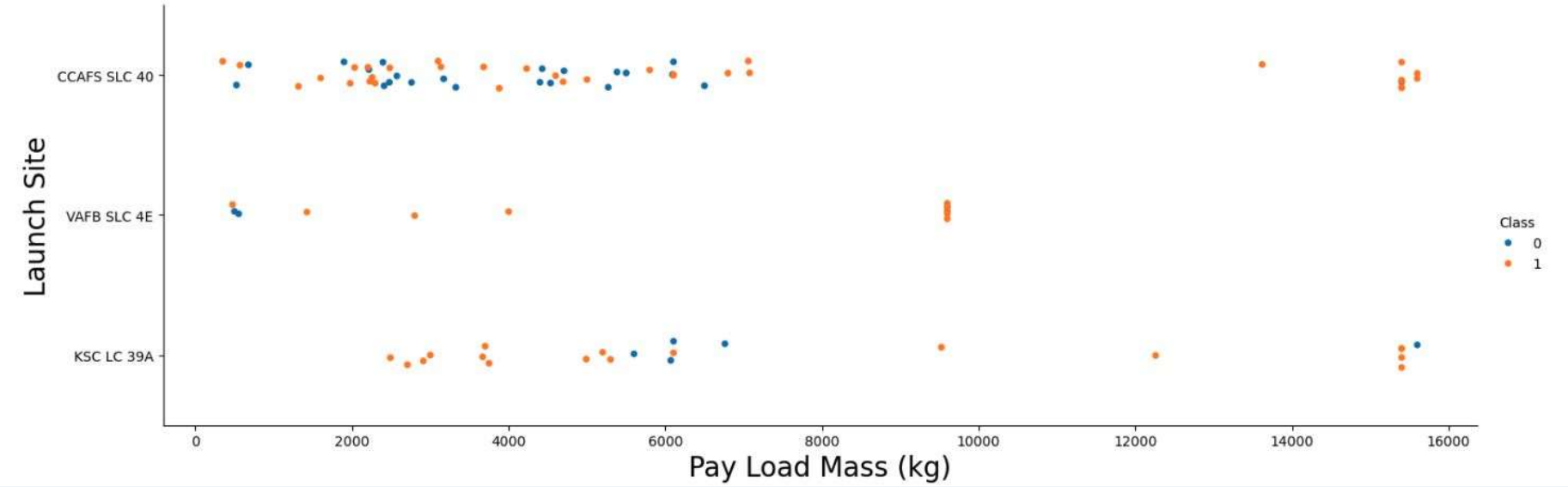
Flight Number vs. Launch Site

- All sites show a mix of first stage landing successes and failures, with successes increasing over time.
- Early flights predominantly resulted in failures, indicating improvements to technology or process.
- While CCAFS SLC 40 has the most total flights, VAFB SLC 4E appears to have a relatively higher.



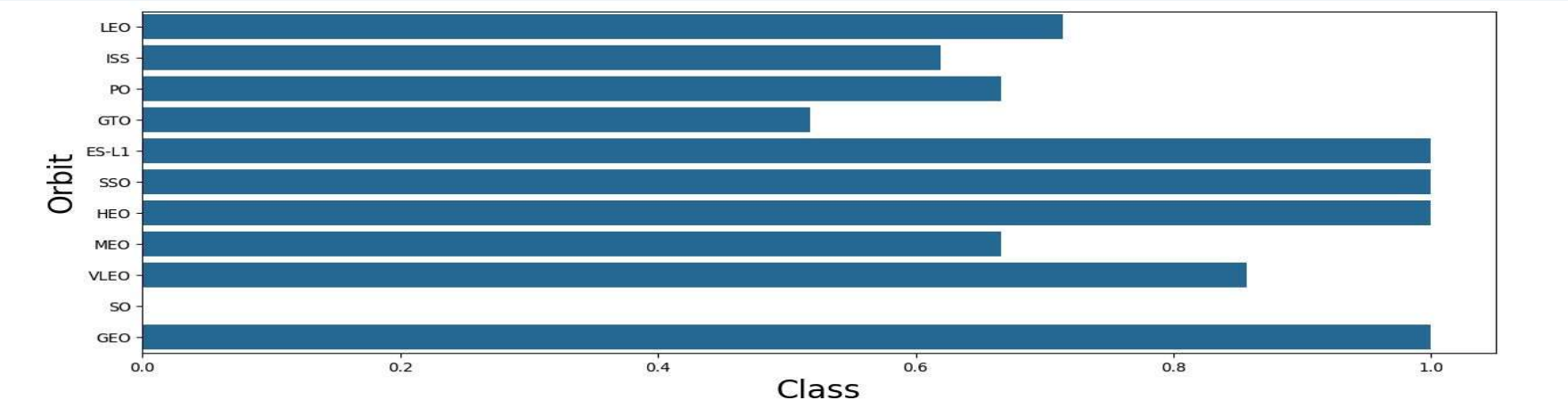
Payload vs. Launch Site

- All sites show a wide range of payload weights, from light to heavy payloads.
- Early flights trend toward lighter payloads, representing the bulk of the landing failures.
- This suggests technological or operational improvements lead to a greater rate of success with heavier payloads.



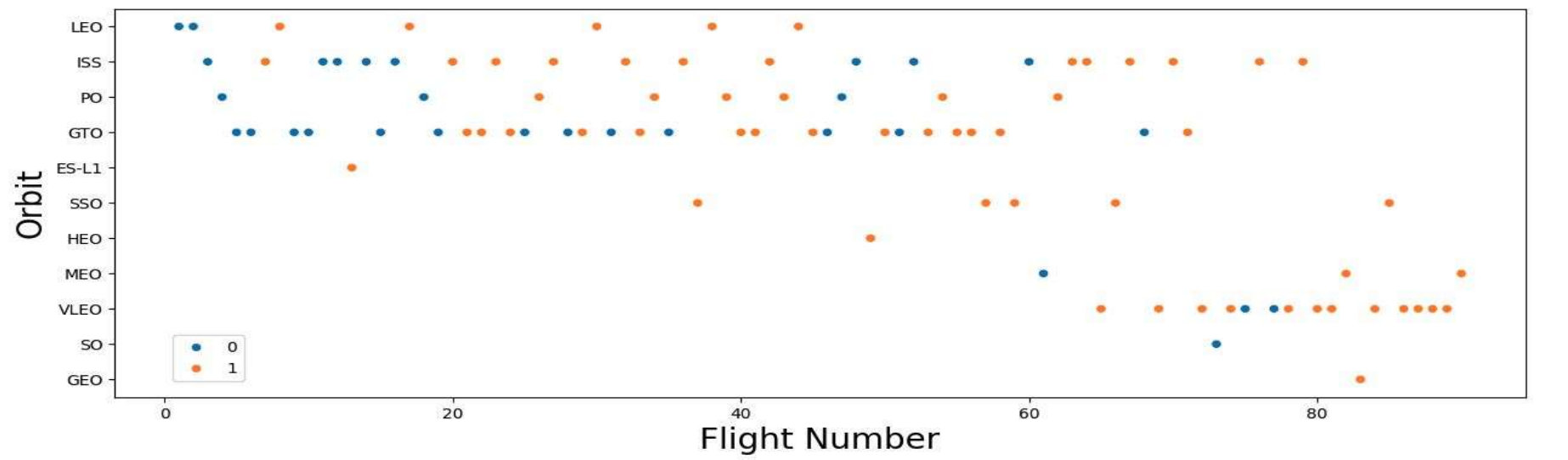
Success Rate vs. Orbit Type

- Some orbits, such as ES-L1, SSO, HEO, and GEO consistently show high success rates.
- Others such as GTO show more mixed outcomes, suggesting some orbit types may introduce operational or technological challenges.
- With only one launch, there is not enough data for the SO orbit type to provide an accurate analysis.



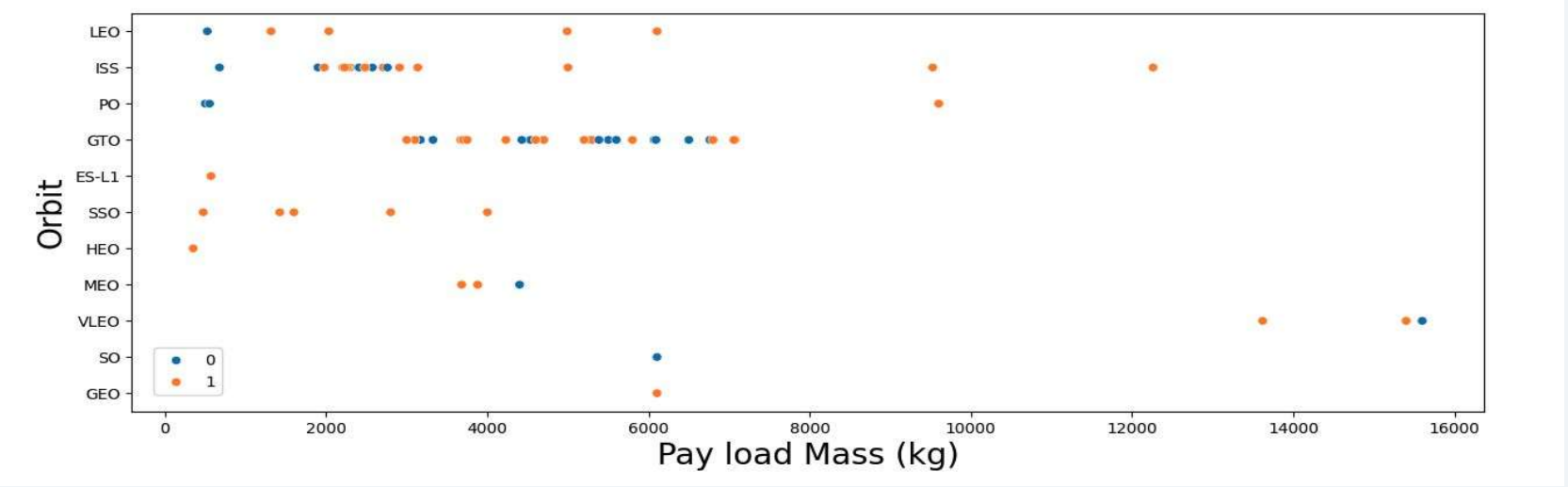
Flight Number vs. Orbit Type

- Many orbits are represented throughout the flight number range, some orbits are not attempted until later flights.
- There is a noticeable improvement in landing success as flight numbers increase, indicating an accumulation of experience and ongoing improvements.



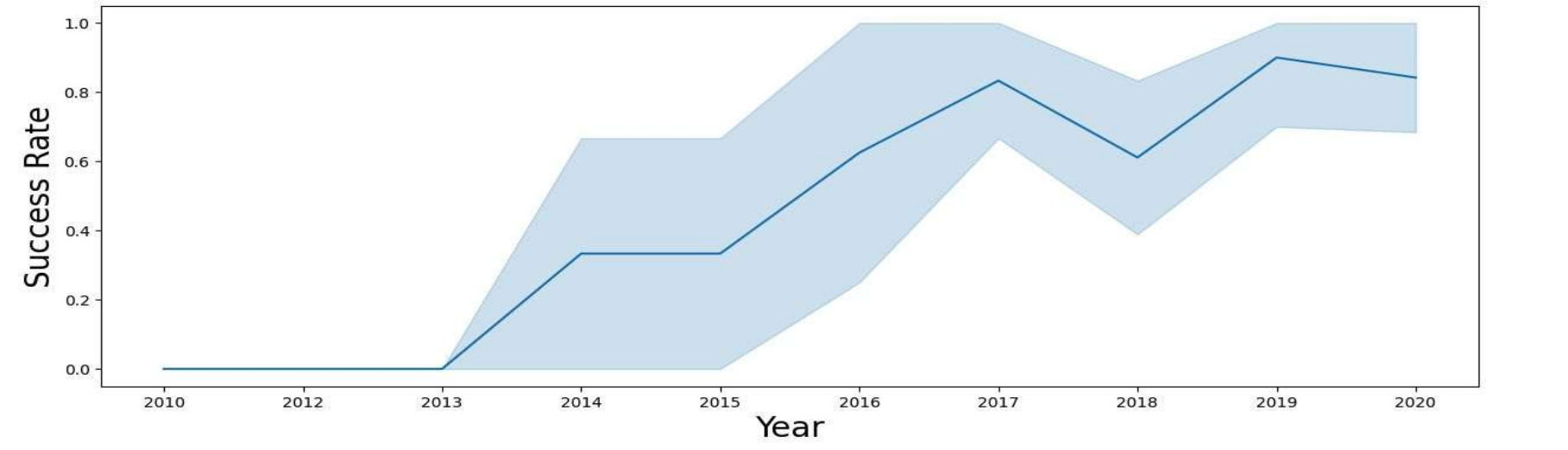
Payload vs. Orbit Type

- Many orbits are represented across a wide range of payload masses, but others like SSO, MEO, HEO and GEO show a generally lower range.
- Orbits with a constrained payload range, tend to show a higher rate of landing success.
- While payload mass does not appear to directly determine mission success, its interplay with orbit suggests a significant correlation.



Launch Success Yearly Trend

- Yearly trend shows a consistent progression from early challenges to high reliability in first stage landings over time.
- From 2016 onward, SpaceX experienced year over year improvement in success rate with a minor setback in 2018.



All Launch Site Names

There are four unique Launch Sites

- o CCAFS LC-40
- o VAFB SLC-4E
- o KSC LC-39A
- o CCAFS SLC-40

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch Site Names Begin with 'CCA'

- First five rows displaying where launch sites begin with `CCA`

```
 * sqlite:///my_data1.db
Done.
```

10]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

Total Payload Mass

The total payload carried by boosters from NASA (CRS) is 45,596kg.

```
SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD
    FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 is 2,534.67kg.

```
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD_MASS
        FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

First Successful Ground Landing Date

The following is a sql query that returns first successful landing outcome on ground pad occurred on December 22nd, 2015:

```
SELECT MIN(Date) as LaunchDate
   FROM SPACEXTABLE
   WHERE Landing_Outcome = 'Success (ground pad)';
```

Successful Drone Ship Landing with Payload between 4000 and 6000

The following are boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

Booster	Payload Mass
F9 FT B1022	4,696kg
F9 FT B1026	4,600kg
F9 FT B1021.2	5,300kg
F9 FT B1031.2	5,200kg

```
SELECT Booster_Version, PAYLOAD_MASS__KG_
   FROM SPACEXTABLE
   WHERE Landing_Outcome = 'Success (drone ship)'
   AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes:

Mission Status	Count
Failure	1
Success	100

```
SELECT CASE
WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
                WHEN Mission_Outcome LIKE 'Failure%' THEN 'Failure'
END as Mission_Status, COUNT(*)
FROM SPACEXTABLE
GROUP BY Mission_Status;
```

Boosters Carried Maximum Payload

- The maximum payload sent was 15,600kg.
- The boosters that carried the maximum payload are:

```
SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_

FROM SPACEXTABLE

WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
ORDER BY Booster_Version;
```

Booster Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

2015 Launch Records

- The following is list of failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:

Month	Outcome	Booster	Launch Site
January	Failure (drone ship)	F9 v1.1 B10112	CCAF5 LC-40
April	Failure (drone ship)	F9 v1.1 B10115	CCAF5 LC-40

```
SELECT
   CASE strftime('%m', Date)
         WHEN '01' THEN 'January'
         WHEN '02' THEN 'February'
         WHEN '03' THEN 'March'
         WHEN '04' THEN 'April'
         WHEN '05' THEN 'May'
          WHEN '06' THEN 'June'
         WHEN '07' THEN 'July'
         WHEN '08' THEN 'August'
         WHEN '09' THEN 'September'
         WHEN '10' THEN 'October'
         WHEN '11' THEN 'November'
         WHEN '12' THEN 'December'
 END as Month,
 Landing_Outcome, Booster_Version, Launch_Site, Date
FROM SPACEXTABLE
WHERE strftime('%Y', Date) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

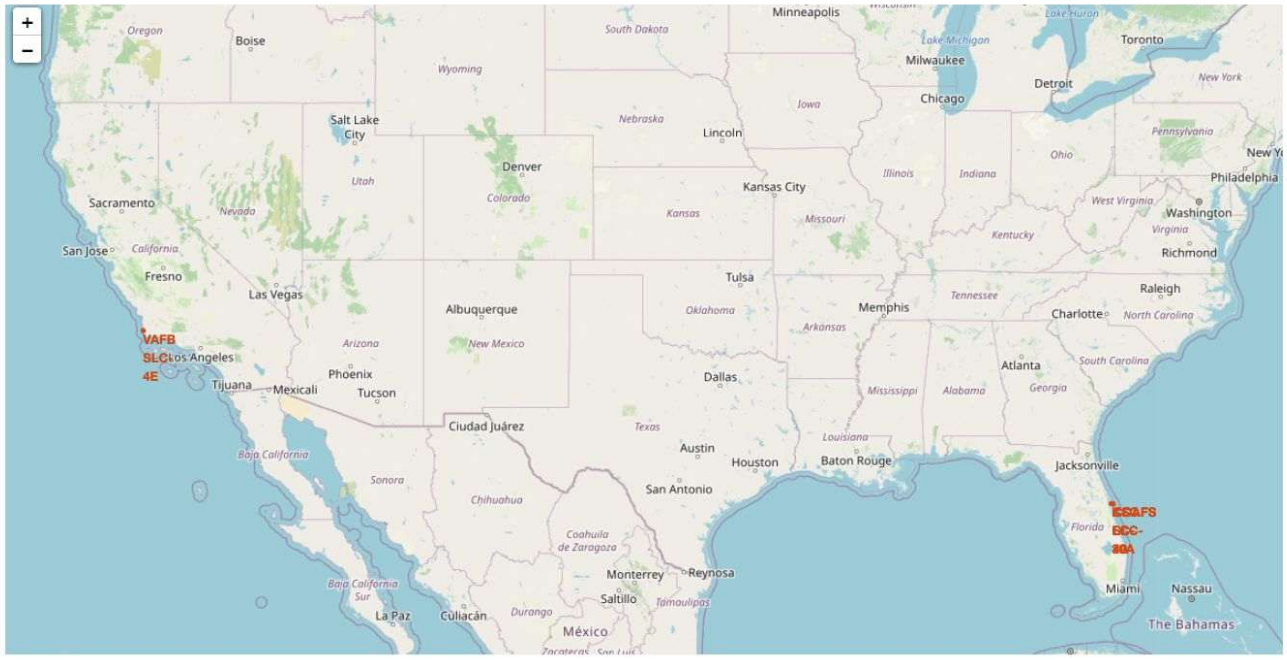
```
SELECT Landing_Outcome, COUNT(*) as Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success(ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Preclude (drone ship)	1

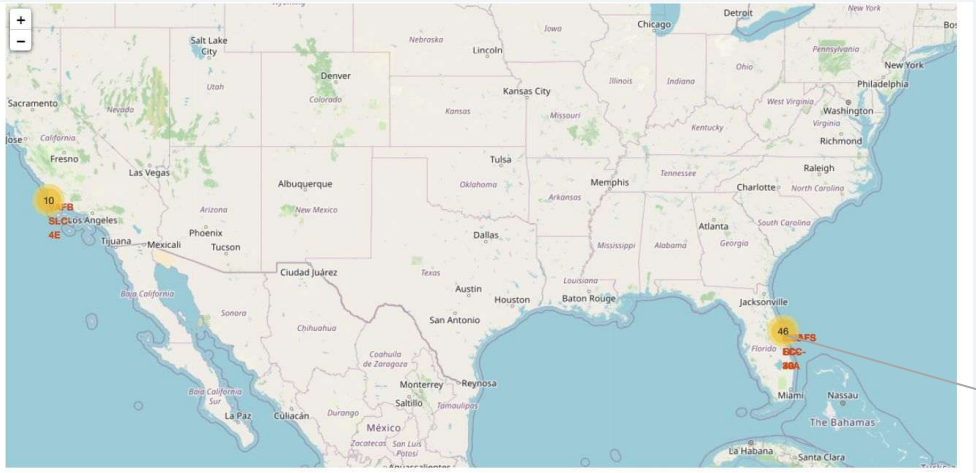

Section 3
Launch Sites
Proximities Analysis

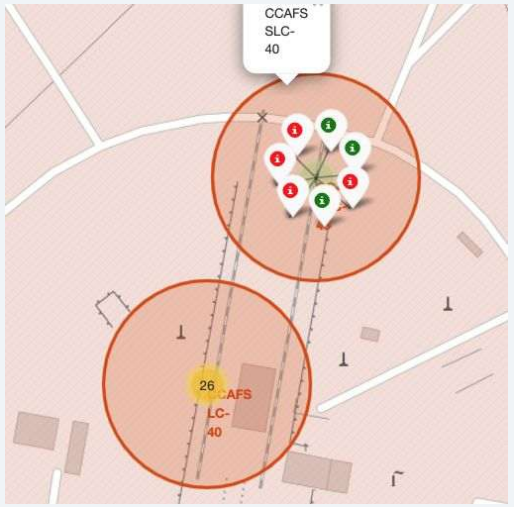
Launch Site Location

- Launch sites are located near coastal region in Florida and California to reduce risk of horrfic failures affecting people.



Launch Outcomes

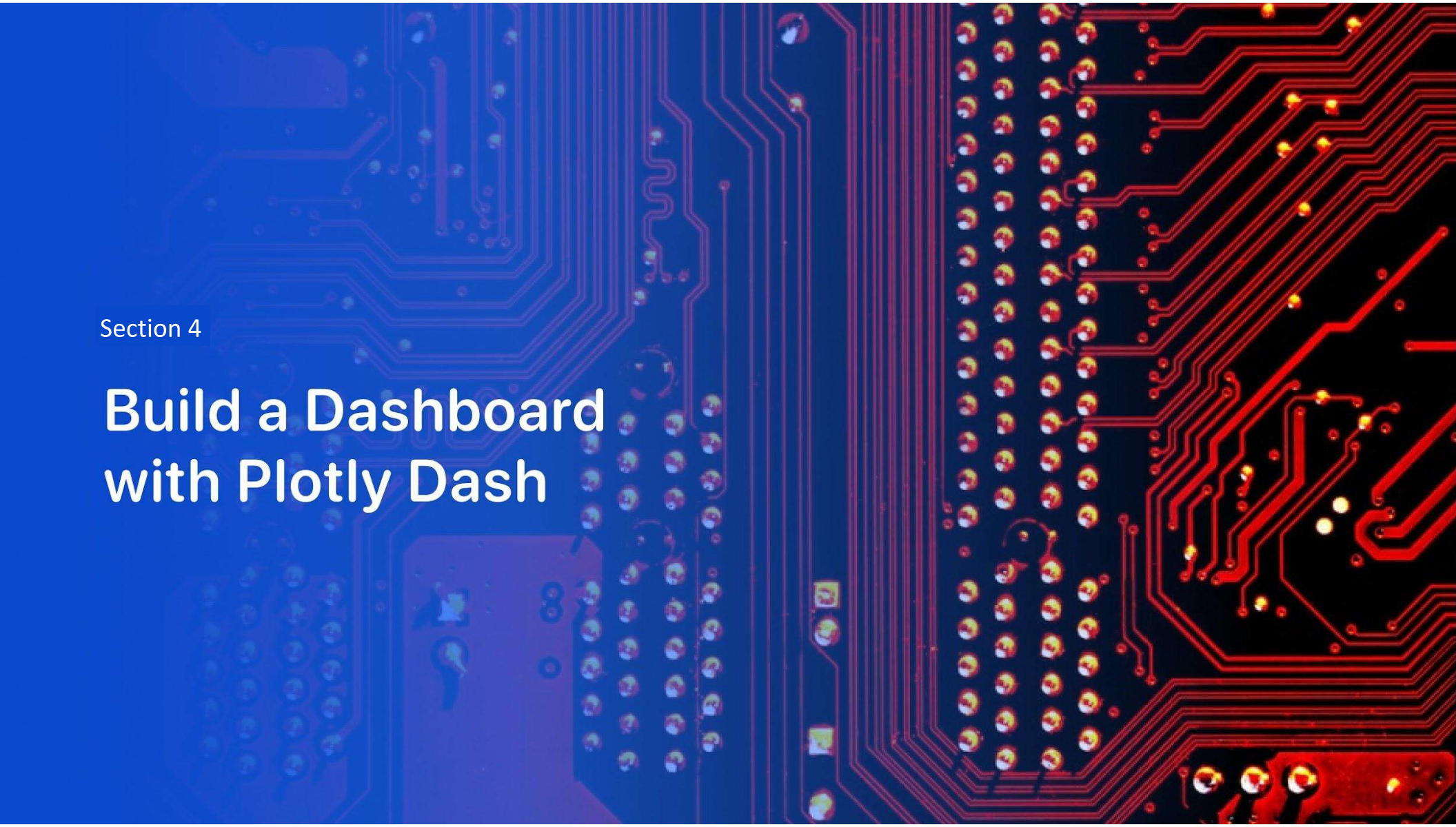




Notable Proximate Locations

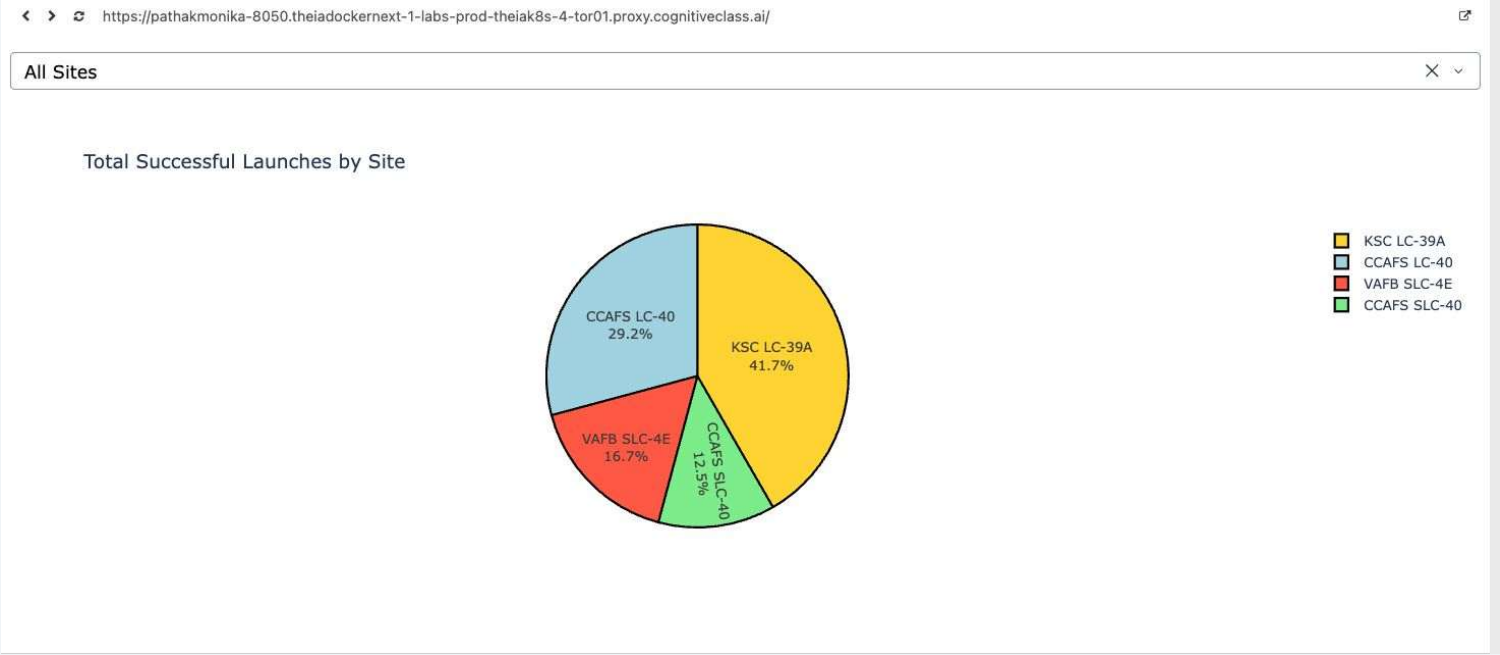
- Notable locations shows
  - Railway (1.02KM)
  - Roadway (0.59KM)
  - Coast (0.86KM)




Section 4
Build a Dashboard with Plotly Dash

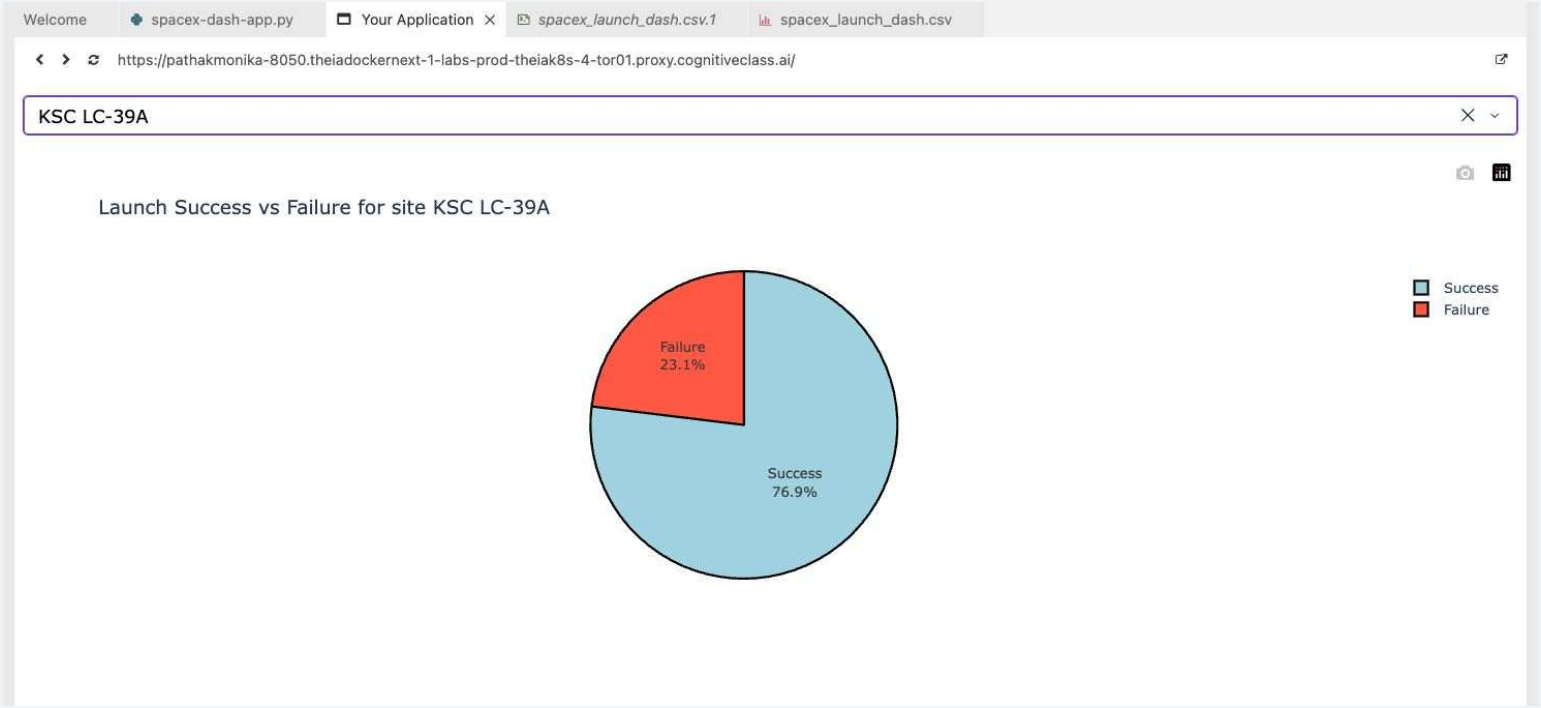
All Launch Sites: Successful Landings

- KSC LC-39A had the highest successful landings, followed by CCAFS LC-40.
- VAFB SLC-4E and CCAFS SLC-40 had the lowest succe



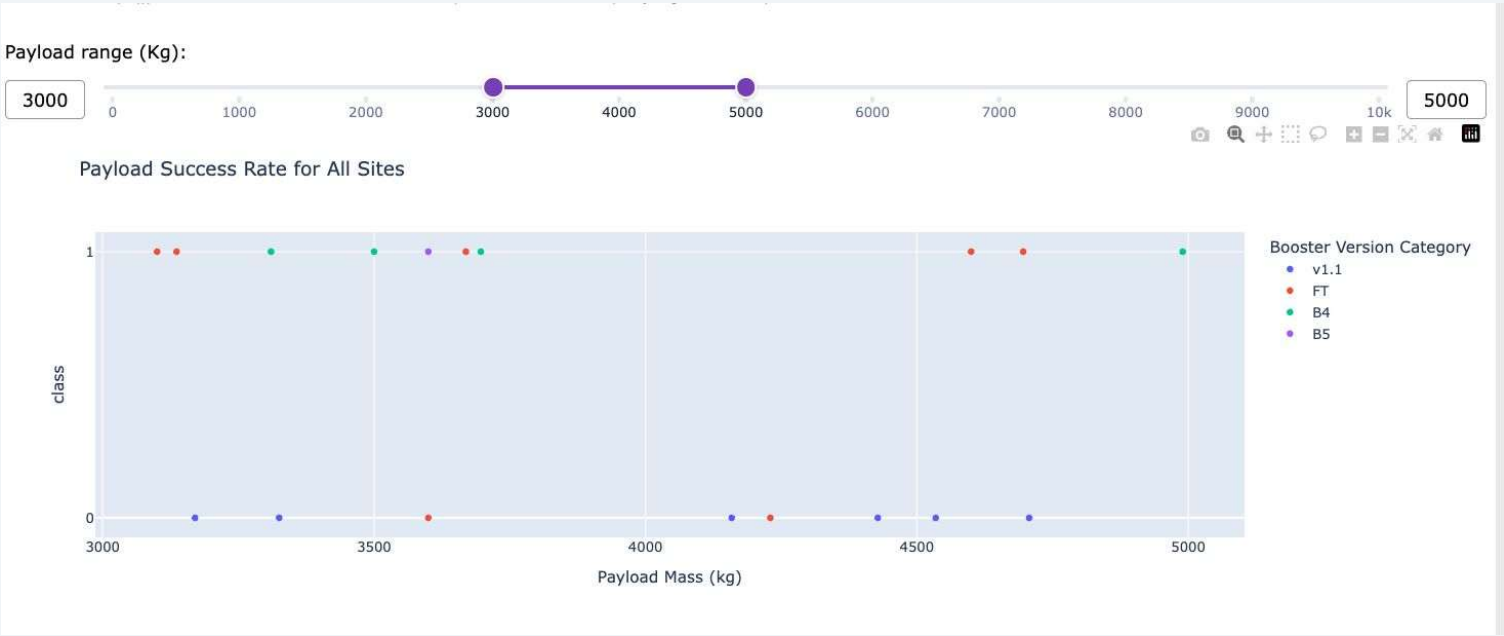
Per-Site Launch Success Ratio: High

- KSC LC-39A launch site had the highest successful landings. Below is the pie graph:



Payload Range

- Payload mass ranging from 3000-5000kg, v1.1 boosters has the worst performance.
- B4 and B5 boosters have the success rate in the payload mass ranging from 3000-5000kg.



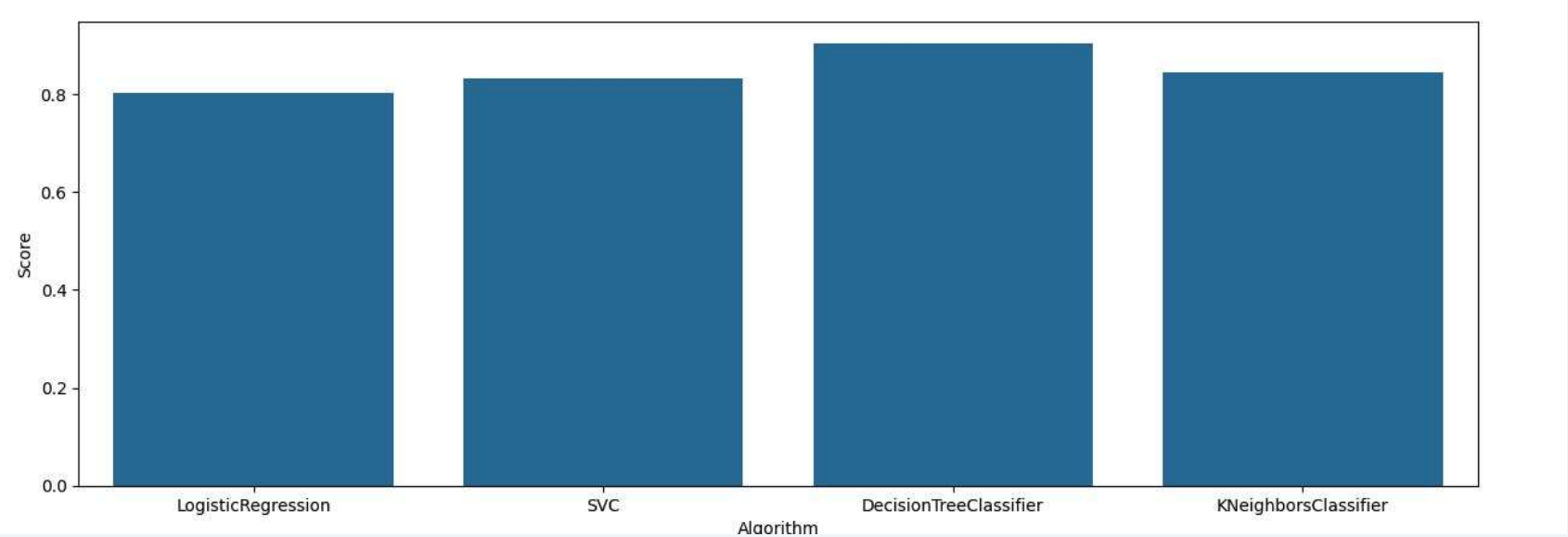
Section 5

Predictive Analysis (Classification)



Classification Accuracy

- Decision Tree Classifier algorithm was the most accurate out of the four algorithm with accuracy test of 0.8333. Below is the bar graph displaying Score vs. Algorithm:



Confusion Matrix

- 14 observations were correctly predicted as successful landings (true positive).
- 3 observations were correctly predicted as failed landings (true negative).
- 1 observation was incorrectly predicted as a successful landing (false positive).
- No observations were incorrectly predicted as a failed landing (false negative).



Conclusions

- Success rates increase over time, across all factors, which indicates continuous and incremental operational improvements and technological advancements.
- Orbits with best success are: ES-L1, SSO, HEO, and GEO.
- Launch site was a highly predictive factor, with KSC LC-39A being a top performer, closely followed by CCAFS LC-40.
- Many of the predictive models evaluated were able to predict landing outcome with an acceptable level of accuracy. In the testing performed, DecisionTreeClassifier produced best results with high accuracy, precision, and recall.

Appendix

- Data Sources
  - SpaceX API : dataset_part_1.csv
  - Wikipedia: List of Falcon 9 and Falcon Heavy launches (June 2021)
  - Data after wrangling: dataset_part_2.csv
  - Geographical data: spacex_launch_geo.csv
  - Interactive data source: space_launch_dash.csv